

gmc2021

Version: 2 (May 3, 2024)

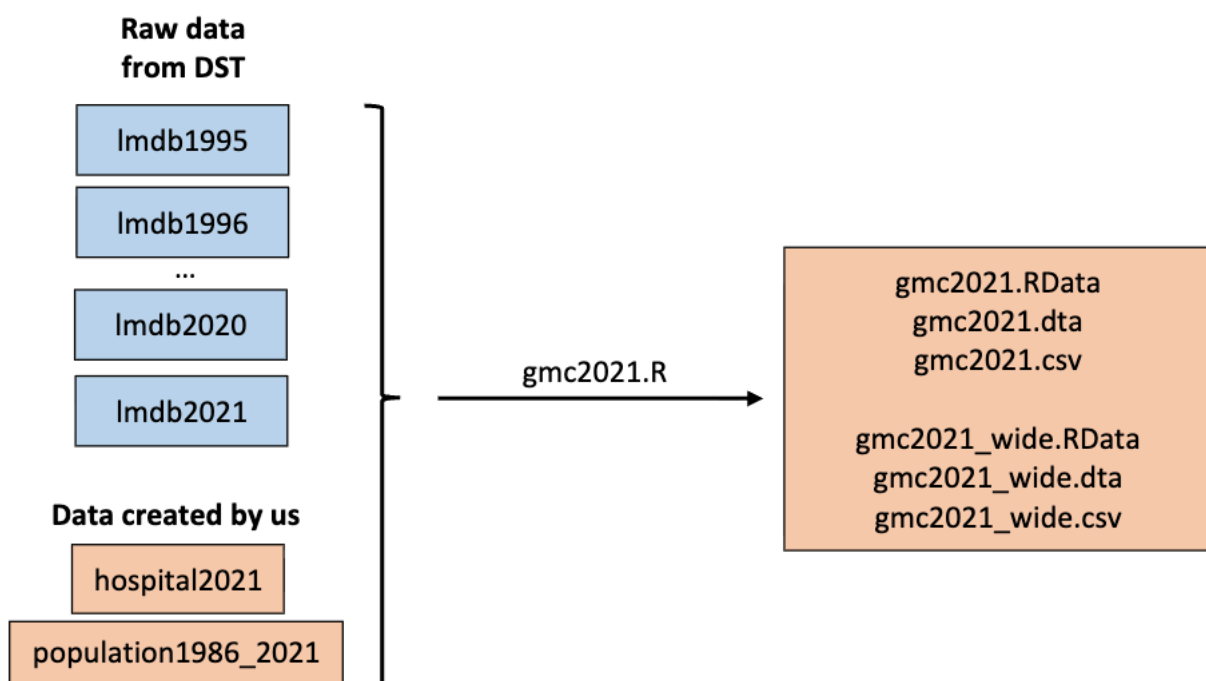
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Changes from previous version: among other ICD-10 codes, we use codes K70-K74 to identify chronic liver disease compared to the previous version that used K70 and K74 only (and not K71-K73).

Additionally, we fixed a problem in the R code that did not consider all cases of hypertension and dyslipidemia.

The datasets `gmc2021` and `gmc2021_wide` contain information on the date of the first appearance in the registers of 39 pre-defined general medical conditions (GMCs) and broad categories of conditions (Table 1). These conditions are identified through diagnoses in Danish hospitals and/or drug prescriptions between January 1, 1995 and December 31, 2021 for all individuals living in Denmark in that period. When these disorders are the outcome of interest, one could consider also identifying them through causes of death (but this is not implemented in this dataset). In `gmc2021`, there is one observation for each person and condition with the date of onset of the condition, while in `gmc2021_wide`, there is one observation for each individual with the date of each of the conditions as additional variables.

Code: `gmc2021.R` (R code)
Output: `gmc2021.RData` and `gmc2021_wide.RData` (R files)
`gmc2021.dta` and `gmc2021_wide.dta` (Stata files)
`gmc2021.csv` and `gmc2021_wide.csv` (CSV files)



The datasets are created based on the following registers:

- Register `population1986_2021`, which includes basic information on all individuals living in Denmark between January 1, 1986, and December 31, 2021. We

have created this dataset based on the registers from [Statistics Denmark \(DST\)](#) on population ([‘Befolkning’; BEF](#)) and deaths in Denmark ([‘Døde i Danmark’; DOD](#)). Specific details and R code are available on OSF (<https://osf.io/zhfyp/>).

- Register `hospital2021`, which includes information on all diagnoses in Danish hospitals (psychiatric and somatic departments). Specific details and R code for register `hospital2021` are available on OSF (<https://osf.io/zhfyp/>). In brief, we have created register `hospital2021` based on (i) the Danish National Patient Register,¹ which contains information on admissions to inpatient facilities since 1977 and visits to outpatient and emergency departments since January 1, 1995; and (ii) the Danish Psychiatric Central Research Register,² which contains information on admissions to psychiatric inpatient facilities since April 1969 and visits to outpatient psychiatric and emergency departments since January 1, 1995. The diagnostic system used in these registers is the ICD-8 from 1969 to 1993, and ICD-10 from 1994 onwards. The register is available through different specific registers from the [Danish Health Data Authority \(Sundhedsdatastyrelsen; SDS\)](#). To create datasets `gmc2021` and `gmc2021_wide`, we considered diagnoses classified as main diagnoses (*hoveddiagnoser/aktionsdiagnoser*; type ‘A’), auxiliary diagnoses (*bidiagnoser*; type ‘B’), and basic diagnoses (*grundmorbus*; type ‘G’). We did not consider associated diagnoses (*tillægsdiagnoser*; type ‘+’), referral diagnoses (*henvisningsdiagnose*; type ‘H’) or diagnoses linked to a hospital contact in which the main diagnosis is related to healthy individuals (i.e., ICD-8 code Y719; ICD-10 codes Z03.2, Z03.8, Z03.9, Z76.3).
- The Danish National Prescription Registry,^{3,4} which contains detailed information on prescriptions redeemed at community pharmacies in Denmark since 1995. There are three notable instances when drugs are not filled at community pharmacies and thus are not captured by the registry: (i) drugs used during hospital admissions; (ii) drugs used by certain institutionalized individuals (typically due to psychiatric illnesses); and (iii) drugs supplied directly by hospitals or treatment centres (e.g. chemotherapeutic agents, immunosuppressant drugs and methadone for those with substance use disorders). The register is available through the Danish Health Data Authority through [Lægemiddelstatistikregisteret](#), or through Statistics Denmark drug database ([‘Lægemiddeldatabasen’; LMDb](#)).

The dataset `gmc2021` contains the date of first appearance for each of the conditions, according to the criteria listed in Table 1. The final dataset contains 32,896,978 observations from 5,232,381 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits’ positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.
- **disease**: Disease identifier as listed in Table 1.
- **disease_label**: Disease label as listed in Table 1.
- **disease_date**: Date of first appearance in the registers of diseases listed in Table 1.

The dataset `gmc2021_wide` contains the date of the first appearance for each of the conditions listed in Table 1 in a wide format. The final dataset contains 5,232,381 observations from 5,232,381 unique identifiers with the following variables:

- **pnr**: Unique personal identifier as defined above.
- **gmc1, gmc2, ..., gmc31**: Date of the first appearance of conditions listed in Table 1.
- **gmc101, gmc102, ..., gmc109**: Date of the first appearance of broad categories listed in Table 1.
- **gmc1101, gmc1102**: Date of the first appearance of sub-disorders listed in Table 1.

Table 1. Classification of diagnosis and/or drug codes into 31 specific general medical conditions and 9 broad categories of conditions. This classification is based on previous publications using Danish registers.^{5,6}

	General Medical Conditions	Coding definition	Diagnosis codes (ICD-10)	Drug codes (ATC) ^a
101	Circulatory system			
1	<i>Hypertension</i>	Diagnosis AND/OR prescriptions of antihypertensives ^b	I10-I13, I15	C02, C03, C04, C07, C08, C09
2	<i>Dyslipidemia</i>	Diagnosis AND/OR prescription of lipid-lowering drugs ^c	E78	C10
3	<i>Ischemic heart disease</i>	Diagnosis AND/OR prescription for antianginal drug	I20-I25	C01DA
4	<i>Atrial fibrillation</i>	Diagnosis	I48	
5	<i>Heart failure</i>	Diagnosis	I50	
6	<i>Peripheral artery occlusive disease</i>	Diagnosis	I70-I74	
7	<i>Stroke</i>	Diagnosis	I60-I64, I69	
102	Endocrine system			
8	<i>Diabetes mellitus</i>	Diagnosis AND/OR prescription of antidiabetics	E10-E14	A10A, A10B
9	<i>Thyroid disorder</i>	Diagnosis AND/OR prescription of thyroid therapy drugs	E00-E05, E061-E069, E07	H03
10	<i>Gout</i>	Diagnosis	E79, M10	
103	Pulmonary system and allergy			
11	<i>Chronic pulmonary disease^d</i>	Diagnosis AND/OR Prescription for obstructive airway disease drugs	J40-J47	R03
1101	<i>Chronic obstructive pulmonary disease</i>	Diagnosis AND/OR Prescription for obstructive airway disease drugs	J40-J44, J47	R03 ^d
1102	<i>Asthma</i>	Diagnosis AND/OR Prescription for obstructive airway disease drugs	J45-J46	R03 ^d
12	<i>Allergy</i>	Diagnosis AND/OR Prescription for non-sedative antihistamines AND/OR nasal antiallergics	J30.1-J30.4, L23, L50.0, T78.0, T78.2, T78.4	R06AX, R06AE07, R06AE09, R01AC, R01AD
104	Gastrointestinal system			
13	<i>Ulcer/chronic gastritis</i>	Diagnosis	K221, K25-K28, K293-K295	
14	<i>Chronic liver disease</i>	Diagnosis	B16-B19, K70-K74, K766, I85	
15	<i>Inflammatory bowel disease</i>	Diagnosis	K50-K51	

16	<i>Diverticular disease of intestine</i>	Diagnosis	K57	
105	Urogenital system			
17	<i>Chronic kidney disease</i>	Diagnosis	N03, N11, N18-N19	
18	<i>Prostate disorders</i>	Diagnosis AND/OR prescription of prostate hyperplasia therapy drugs	N40	C02CA, G04C
106	Musculoskeletal system			
19	<i>Connective tissue disorders</i>	Diagnosis	M05-M06, M08-M09, M30-M36, D86	
20	<i>Osteoporosis</i>	Diagnosis AND/OR prescription for osteoporosis drugs	M80-M82	M05B, G03XC01, H05AA
21	<i>Painful condition</i>	Repeated prescriptions of analgesics		N02A, N02BA51, N02BE, M01A, M02A
107	Hematological system			
22	<i>HIV/AIDS</i>	Diagnosis	B20-B24	
23	<i>Anemias</i>	Diagnosis	D50-D53, D55-D59, D60-D61, D63-D64	
108	Cancers			
24	<i>Cancers</i>	Diagnosis	C00-C43, C45-C97	
109	Neurological system			
25	<i>Vision problem</i>	Diagnosis	H40, H25, H54	
26	<i>Hearing problem</i>	Diagnosis	H90-H91, H931	
27	<i>Migraine</i>	Diagnosis AND/OR prescription of specific anti-migraine drugs	G43	N02C
28	<i>Epilepsy</i>	Diagnosis AND prescription of anti-epileptics	G40-G41	N03
29	<i>Parkinson's disease</i>	Diagnosis	G20-G22	
30	<i>Multiple sclerosis</i>	Diagnosis	G35	
31	<i>Neuropathies</i>	Diagnosis	G50-G64	

^aWithin each medical condition, prescriptions were only considered if there were at least two prescriptions within a year. For painful condition (21), the requirement was four prescriptions within a year.

^bPrescriptions were used to identify hypertension (1) only if the person did not experience a previous ischemic heart disease (3) or heart failure (5). Additionally, diuretic drug prescriptions (ATC code C03) were used only if the person did not experience a previous chronic kidney disease (17).

^cPrescriptions were used to identify dyslipidemia (2) only if the person did not experience a previous ischemic heart disease (3).

^dChronic pulmonary disease (11) was additionally subdivided into chronic obstructive pulmonary disease (COPD; 1101) and asthma (1102). Due to the lack of specificity of the medications used to ascertain chronic pulmonary disease, the age at prescription was used. For ATC code R03, the second prescription within a year was assumed to be for asthma if the age at prescription was below the median age of all R03 prescriptions, and for COPD if it was equal to or above the median age of all R03 prescriptions.

1. Lynge E, Sandegaard JL, Rebolj M. The Danish National Patient Register. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):30-33. doi:10.1177/1403494811401482

2. Mors O, Perto GP, Mortensen PB. The Danish Psychiatric Central Research Register. *Scand J Public Health*. 2011;39(7 Suppl):54-57. doi:10.1177/1403494810395825

3. Wallach Kildemoes H, Toft Sorensen H, Hallas J. The Danish National Prescription Registry. - PubMed - NCBI. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):38-41. doi:10.1177/1403494810394717
4. Pottegård A, Schmidt SAJ, Wallach-Kildemoes H, Sørensen HT, Hallas J, Schmidt M. Data Resource Profile: The Danish National Prescription Registry. *Int J Epidemiol*. Published online October 27, 2016:dyw213. doi:10.1093/ije/dyw213
5. Momen NC, Plana-Ripoll O, Agerbo E, et al. Association between Mental Disorders and Subsequent Medical Conditions. *New England Journal of Medicine*. 2020;382(18):1721-1731. doi:10.1056/NEJMoa1915784
6. Prior A, Fenger-Grøn M, Larsen KK, et al. The Association Between Perceived Stress and Mortality Among People With Multimorbidity: A Prospective Population-Based Cohort Study. *American Journal of Epidemiology*. 2016;184(3):199-210. doi:10.1093/aje/kwv324

